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THE TABLE OF ELEMENTAL AFFINITIES.

By Metcalfe, of Kanaich.

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Communicated to the Silver Society, C.R. 1271.

How near a material comes to each of the four elements, fire, water, air and earth, is called its affinity. When Latham classed the monsters more than two hundred years ago, affinity could be told only by the caster's own sense of which element came first and which after; it was not until the craftsmen made the affinity meter that affinity became a number that could be measured, and not until it had been measured again in several cities and upon several instruments, and found to agree, that the learned would use it. At Kanaich I have spent thirty years in measuring the affinities of more than eleven hundred kinds of materials and crystals. I shall show four things in this paper.

First, the affinities of natural crystals of a single form all fall upon a set of points, each of the four affinities being so many sixths; the affinities of twinned crystals fall upon the line between two such points, and those of other materials anywhere. Secondly, by the nomological algebra of Elworthy together with trials of added bias, it can be reckoned which of these points can form a crystal: at four of them the lines in the lattice cancel one another and no crystal can form; at four more the lattice brings forth a twist greater than an ordinary place can bear, so that no crystal can stand there in ordinary places; at the other seventy-six a crystal can form in ordinary places, and all seventy-six I have found. Thirdly, crystals that stand in one cell of the table have the same store and the same form, while their hardness depends on which pairs of lines cancel in their lattice. By these rules the eighty-four points are laid out in a table. Fourthly, the four points that cannot stand I leave as empty cells in the table; their affinity, store, hardness and form can be written down beforehand.

I. THE AFFINITY METER AND ITS ERROR

The meter I use is made after the common pattern. The sample is set in the middle of four small arrays, which stir fire, water, air and earth in turn, and the response of the sample to each is read once; from the four readings the background of the place is first taken off, and each is then divided by the sum of the four, and the four parts so found are the affinity. Before every batch I set the four readings right upon one standard sample; every sample is measured twice, the second time with the order in which the four arrays are stirred reversed, and the mean of the two is taken, lest the first stirring draw the second aside. While the measuring goes on, no casting is allowed within thirty paces of the room. In tide years the background rises and falls so that it cannot be cleanly taken off, and in the three tide years 1250, 1260 and 1270 I measured nothing, but only put the old records in order. All the affinities

here spoken of were measured at the common warmth of a room, in an ordinary place and in ordinary years.

When one sample is measured ten times over, the spread of each of the four parts about its mean (the square root of the mean of the squared differences) is no more than 0.004. I also sent a hundred samples to Onais, and had each measured upon a meter there; the two places differ by no more than 0.006. Only when all four parts lie within 0.012 of those of some point, which is three times the spread, do I say that an affinity falls upon that point. This test is severe: a point taken at random within the tetrahedron has less than two chances in a thousand of passing it.

The materials and crystals I have measured in thirty years fall into three classes, set out in Table I. By a single form I mean that a crystal is of one form throughout, so that neither in cutting it nor under the lens can two crystals be seen grown together; every crystal was judged so before its affinity was measured, and none was classed after its affinity was seen.

TABLE I. The materials and crystals measured in thirty years

Class	*Kinds*	*Pieces*	*Where the affinities fall*
Natural crystals of a single form	76	650	All upon the points, within 0.012
Twinned and mixed crystals	178	403	Twinned upon the line between two points; mixed within the space bounded by several points
Other materials (metals, stones, woods, bones, horn, glazes, alloys and the like)	862	1,938	Anywhere
Total	1,116	2,991	

II. THE TETRAHEDRON

Since the four affinities together always make one, every sample can be drawn as a point within a regular tetrahedron: the four corners are the four pure elements, and how far a point lies from the face opposite a corner tells what part that element has. Every point within the tetrahedron is a possible affinity.

When the two thousand nine hundred and more pieces are drawn in, the three classes look altogether different. The other materials are scattered through every part of the tetrahedron, without rule; among the nineteen hundred and more, three happen to fall upon points, two metals and a glaze, none of them crystals, and by the chance just given some three or four such were to be expected. The affinity of a twinned crystal always falls upon the line joining some two of the points, and nearer to one end or the other according to how much of each crystal it holds: I cut open forty of them and measured the room each of the two crystals took up, and it agreed with the place upon the line within three in a hundred of the room. It follows that the response of the several crystals to the four stirrings, taken together, differs little from one to another for the same room; otherwise the place of a twin would be divided by the responses and not by the room. A mixed crystal is several crystals

grown together, and its affinity lies within the space bounded by those points. With the natural crystals of a single form it is otherwise: of six hundred and fifty pieces, there is not one that does not fall upon a point. At these points each of the four affinities is so many sixths. I write them as sixths of fire, water, air and earth, in that order; thus a crystal four sixths fire and two sixths air is written $(4, 0, 2, 0)$. There are eighty-four such points within the tetrahedron, as in Fig. 1.

Why the points should lie a sixth apart I cannot say. I suppose that every lattice-cell of a crystal holds six strands of line, each strand belonging to one element, and that the affinity is the proportion in which the elements hold strands. This is only a supposition, and the reckonings of the third to the sixth sections all rest upon it. Whether it stands must be judged from the bias of Table II, above all from the twelve kinds whose frame lies upon an axis though they have lines of both axes, and from whether the hardness of Table V goes by steps according to the strands.

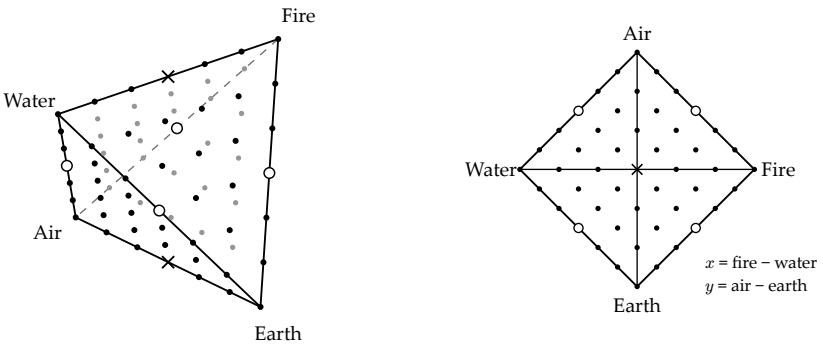


Fig. 1. The tetrahedron of affinities: the four corners are the four pure elements, and the points lie a sixth apart (only the edges, and the points upon the faces, are drawn; a cross marks a point whose frame is null, an open circle an empty cell). Taking fire less water across and air less earth upward, and so casting the tetrahedron upon a plane, gives the table of Fig. 2

III. THE LINES IN THE LATTICE

A crystal is a crystal because the lines of law in its lattice are woven into a frame that can repeat itself. Whether a crystal can form at a given point depends on whether the lines of that point can be woven into such a frame; and the nomological algebra of Elworthy serves exactly for this.

I asked two law-mages of Kanaich to measure, after Elworthy's manner, the lines in each of the four pure crystals. The line in the crystal of fire I write F; that in the crystal of air, G. The line in the crystal of water rises and falls exactly contrary to that of fire, and is the reverse of F, that is $-F$; that in the crystal of earth is $-G$. When F and G are applied in succession they behave as Elworthy's weft and warp: F first and G after raises a twist upon the web that belongs neither to fire and water nor to air and earth, and the contrary order raises the contrary twist. This twist is of the same kind as Elworthy's line of bias, and I call it bias too, and write it S:

$$GF = S, \quad FG = -S, \quad FF = GG = -1 \tag{1}$$

The successive applications of F, G and S go exactly as those of the weft, the warp and the bias in Elworthy's first table, so that his equation (7) can be used as well.

If the lattice-cell at a point (f, w, a, e) is woven of six strands, as supposed above, its frame is the joint application of the six:

$$L = fF + w(-F) + aG + e(-G) = xF + yG, \quad x = f - w, \quad y = a - e \tag{2}$$

Here x is fire less water and y is air less earth. The lines of fire and water cancel in pairs, and so do those of air and earth; what cancels does not enter the frame, but it is still in the lattice. In one lattice-cell there are $\min(f, w)$ cancelled pairs of fire and water and $\min(a, e)$ of air and earth. By Elworthy's equation (7) the strength of the frame is

$$|L|^2 = x^2 + y^2 \tag{3}$$

From (2) the first thing is plain at once: when x and y are both nought the frame is the null, and nothing can be woven. There are only four points at which f equals w and a equals e: $(3, 3, 0, 0)$, $(0, 0, 3, 3)$, $(2, 2, 1, 1)$ and $(1, 1, 2, 2)$, which stand together in the middle cell of the table of the fourth section. At the point where fire and water have half each, the lines of fire and of water in the lattice cancel exactly; it is the same at the point of half air and half earth; and at the other two all four lines cancel in pairs. Among the six hundred and fifty natural crystals of a single form there is not one at any of these four points. An even twin may also fall upon these points, as a twin of pure fire crystal and pure water crystal in equal parts falls upon $(3, 3, 0, 0)$; and so only pieces cut open and found of one form throughout are counted here.

The second thing is the bias. Where strands of one axis cross, by (1) they yield only multiples of the identity operation, and no bias. Where a strand of the axis of fire and water crosses a strand of the axis of air and earth, every crossing brings forth by (1) one share of bias, either forward or backward. In a lattice-cell every strand of the one axis crosses every strand of the other once; and if at every crossing the axis of fire and water comes first, the bias brought forth in one lattice-cell is

$$(f - w)(a - e)S = xyS \tag{4}$$

I call $b = xy$ the number of bias of that point, the bias being counted in shares of S, one share to a crossing. Whether the two axes always cross in this order can only be measured. I asked the same two law-mages to measure the bias in every one of the fifty-two crystals whose frame has both x and y other than nought, without telling them the affinities of the pieces; two law-mages of Onais measured twenty of them again, and agreed. The results are in Table II: in every one bias could be measured, and its number agreed with xy in size and, in every kind, in sign. Agreement in size shows only that within one crystal the order of crossing is fixed; agreement in sign shows that the axis of fire and water comes first. In the twenty-four crystals whose frame lies upon one axis, the pure crystals among them, no bias could be measured; and twelve of these have lines of both axes, those of the other axis cancelling in pairs, as in $(2, 0, 2, 2)$, where a lattice-cell has eight crossings, four forward and four

backward, which come to nothing, as (4) requires. Why the axis of fire and water should come first, I do not know.

Table II. The number of bias measured in fifty-two crystals (error of measuring bias some five in a hundred)

Size reckoned by (4)	*Kinds*	*Mean measured*	*Least*	*Greatest*	*Kinds agreeing in sign*
1	12	1.01	0.97	1.05	12
3	16	3.01	2.80	3.20	16
4	8	3.97	3.77	4.18	8
5	8	4.94	4.64	5.18	8
8	8	8.09	7.70	8.76	8

Elworthy conjectured that in the old wards of two lines the share of bias brought forth by the warp and weft in succession has nowhere to go, and that when enough of it gathers the ward tears there. The bias in a lattice must also go somewhere. There is no line of bias in the lattice, and the bias can only be borne off by the laws of the place about it; this is my supposition. By it, how much bias a place can bear belongs to the place, and can be tried.

I asked the law-mages to make trials of added bias in the trial room at Kanaich, all in ordinary years: a crystal is set in an array, and bias is applied to it step by step, each step a tenth of a share, in the same direction as the bias it has of its own, until the crystal cracks. Seven kinds were tried with three pieces each, the pieces of different sizes, from a tenth of an inch to an inch, as in Table III. Those with much bias of their own crack after a few steps; those with little need many; but what they had of their own and what was added, taken together, comes in all twenty-one pieces to some eight shares and a half, the least 8.4 and the greatest 8.7, whatever the size of the piece. I also added bias the contrary way: $(6, 0, 0, 0)$, which has no bias of its own, cracked again at some eight and a half; $(5, 0, 1, 0)$, which has five shares forward of its own, needed thirteen and a half added backward before it cracked, which together come to eight and a half backward. The bound is the same either way, and the bias goes by forward and backward together; this is a further support. Two kinds tried again at Onais gave the same. By the supposition above, this number should belong to the place; Table III shows that it does not change with the crystal, with the direction, or with the city, which agrees.

TABLE III. Trials of added bias (each step a tenth of a share; the bias of its own reckoned by (4); the total is that bias together with what was added when it cracked, backward counted negative)

Crystal	*Own bias*	*Added at cracking, three pieces*	*Total*
$(6, 0, 0, 0)$	0	8.5, 8.6, 8.6	8.5–8.6
$(2, 0, 2, 2)$	0	8.6, 8.7, 8.6	8.6–8.7
$(2, 1, 2, 1)$	1	7.6, 7.5, 7.5	8.5–8.6
$(4, 1, 1, 0)$	3	5.5, 5.6, 5.6	8.5–8.6
$(3, 1, 2, 0)$	4	4.6, 4.4, 4.6	8.4–8.6
$(5, 0, 1, 0)$	5	3.5, 3.5, 3.6	8.5–8.6
$(4, 0, 2, 0)$	8	0.7, 0.6, 0.5	8.5–8.7
$(6, 0, 0, 0)$, backward	0	8.5, 8.6, 8.5	−8.5 to −8.6
$(5, 0, 1, 0)$, backward	5	13.5, 13.5, 13.5	−8.5
$(4, 1, 1, 0)$, Onais	3	5.6, 5.6, 5.6	8.6
$(4, 0, 2, 0)$, Onais	8	0.6, 0.6, 0.6	8.6

The third thing is now plain as well. The greatest size of xy among the eighty-four points is nine, and it belongs to exactly four points, those of half fire and half air, half fire and half earth, half water and half air, and half water and half earth: $(3, 0, 3, 0)$, $(3, 0, 0, 3)$, $(0, 3, 3, 0)$ and $(0, 3, 0, 3)$. At these four the frame is not the null and the two kinds of line are compatible (since crystals with lines of both axes grow, the four lines are compatible in a lattice), so that by the nomological algebra a frame can be woven; but one lattice-cell brings forth nine shares of bias, and an ordinary place can bear only eight and a half, so that in ordinary places they cannot stand. The eight points whose number of bias is eight fall short of this bound by only half a share; the greatest measured in Table II, 8.76, is within the error of measuring bias. Of these eight I shall speak again in the sixth section. At the other seventy-six points the frame is not the null and the bias is no more than eight, so that they can all form crystals in ordinary places; and these seventy-six crystals I have all found.

IV. THE TABLE OF AFFINITIES

A tetrahedron is not easily drawn on paper, and I have cast it upon a plane, taking fire less water, that is x, across, and air less earth, that is y, upward. The four corners of fire, water, air and earth then fall at the four corners of a square; the edge of fire and water falls upon the axis across, that of air and earth upon the axis upward, and the other four edges are the four sides of the square. The eighty-four points fall into forty-nine cells: the one cell in the middle holds the four points at which x and y are both nought, and none of the four can form a crystal; each of the eight cells nearest the middle holds three points; each of the sixteen in the next ring holds two; and each of the twenty-four in the outermost ring holds one, the four at the middles of the sides being the empty cells. The several kinds in one cell have the same frame and the same number of cancelled pairs, and differ only in which pairs cancel: one has a pair of fire and water more, another a pair of air and earth more. This table I call the table of elemental affinities; it is drawn in Fig. 2.

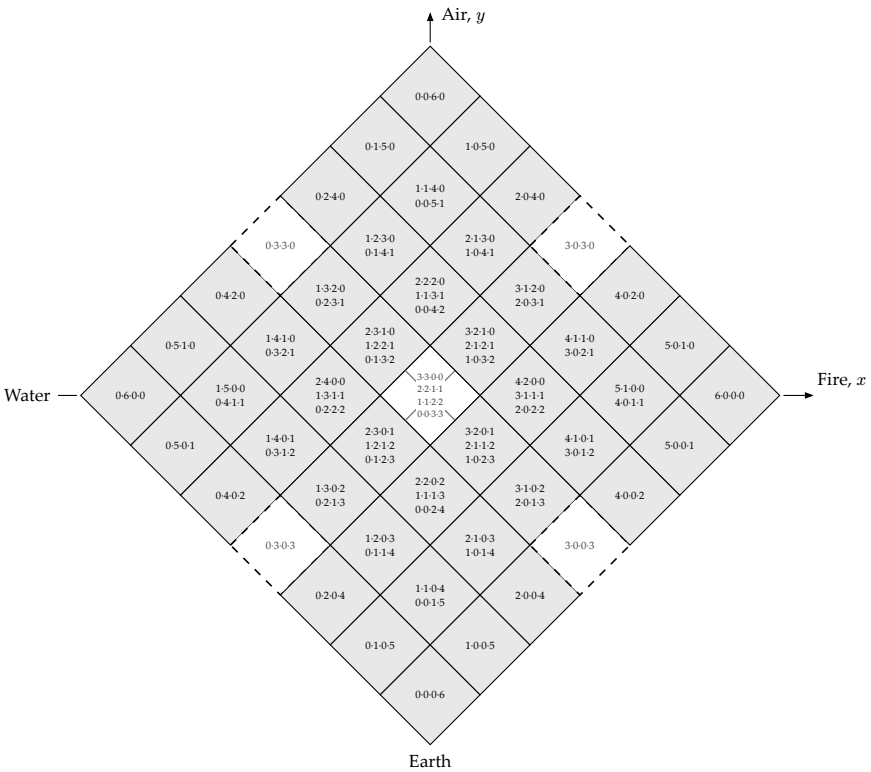


Fig. 2. The table of elemental affinities. Across, fire less water; upward, air less earth; in each cell the affinities of the points it holds (fire, water, air, earth, in sixths). The filled cells have all been found as natural crystals; the crossed cell in the middle has a null frame and can form no crystal; the four open cells at the middles of the sides are the empty cells, whose bias passes the bound an ordinary place can bear

From this table it appears that most of the qualities of a crystal are fixed by the cell in which it stands: the cell fixes its frame and its bias, and its place within the cell fixes which pairs cancel in it. The next two sections speak of the store, the form and the hardness.

V. THE STORE AND THE FORM

The store is measured by Finch's rules: the crystal is filled, joined to the discharging apparatus, and discharged down to the prescribed floor, and what it gives is its store, in the Society's units of exchange, divided by the room the crystal takes up, so many to the cubic inch; the room is found by burying the crystal in fine sand and measuring how far the sand rises. A crystal discharged to the floor is not spoiled, and may be filled again after it has been measured; what stays below the floor differs from kind to kind by no more than a hundredth of the store. Of each of the seventy-six kinds at least two pieces were measured, and the mean taken. The eight kinds whose number of bias is eight are all small, a grain storing only a few

hundredths of a unit, while one discharge of Finch's apparatus has an error of some tenth of a unit, so that a single grain cannot be measured; of these eight I filled and discharged some tens of grains of one kind together, and divided by their room taken together, with an error of some four in a hundred.

I have grouped the seventy-six by the strength of their frame by (3), as in Table IV. In each group the frames point in several ways, and in the groups that have cancelled pairs, which pairs cancel differs from one kind to the next; yet the store to the cubic inch of every kind lies within five in a hundred of the mean of its group; and the stores of the groups stand in proportion to the strength of the frame, each share of strength storing some 0.20 unit. Between the groups it is plainer still: the groups of strength two and four, both with two cancelled pairs, differ in store by one to two; the groups of strength four, sixteen and thirty-six, all without bias, store 0.80, 3.21 and 7.19, in proportion to their strength. The four pure crystals have the strongest frames and the greatest stores; the eight cells about the middle have frames of only two or four shares of strength, and their stores are small. The lines that cancel do not enter the frame, and it seems that they hold no mana either.

TABLE IV. The store to the cubic inch of the seventy-six crystals, grouped by the strength of the frame (units of exchange)

Strength of frame	*Kinds*	*Mean store*	*Least*	*Greatest*	*0.20 times strength*
2	12	0.40	0.39	0.41	0.40
4	12	0.80	0.78	0.82	0.80
8	8	1.59	1.56	1.64	1.60
10	16	2.01	1.92	2.08	2.00
16	8	3.21	3.12	3.28	3.20
20	8	3.94	3.83	4.01	4.00
26	8	5.25	5.13	5.37	5.20
36	4	7.19	6.95	7.39	7.20

The form, too, goes by the cell. The several kinds in one cell all have the same form; and in different cells, those whose frames point the same way have the same form: in the cells upon the axis across the crystals are prisms of six sides, upon the axis upward they are thin plates, upon the two slanting lines that run through the four empty cells, where fire less water equals air less earth or its contrary, double pyramids, and in all the rest slanting prisms, the slant of the faces turning with the way the frame points. I first classed the crystals by their forms, and learned only afterwards that the form follows the way the frame points alone, and not its strength, nor what cancels.

VI. THE HARDNESS AND THE BIAS

The hardness is measured by the scoring needle of the Kanaich workshops: a needle under a fixed weight is drawn across a face of the crystal, the breadth of the score is measured, and it is turned into a number upon a scale of ten; the greater the number, the harder the crystal. The kinds whose number of bias is eight are too

small and too brittle for this, and were scored with a finer needle under half the weight, and the scores turned into the same scale, with a larger error of some two tenths. The hardness of the seventy-six goes only with the cancelled pairs, as in Table V. The number of cancelled pairs is fixed by the frame, so that the groups differ in the strength of their frames as well; but within one group the hardness does not change with the frame or the bias: among the twenty kinds without cancelled pairs, the four whose number of bias is nought have a mean of 3.20, the eight whose number is five 3.19, and the eight whose number is eight 3.18. The kinds without cancelled pairs are all near three; every cancelled pair of fire and water adds some 1.8 to the hardness, and every pair of air and earth some 1.4. Thus the several kinds within one cell have the same store, but their hardness stands in steps: the one with a pair of fire and water more is harder than the one with a pair of air and earth more. The lines that cancel do not enter the frame, but they bind the lattice tight.

Table V. The hardness of the seventy-six crystals, grouped by their cancelled pairs

Pairs of fire and water	*Pairs of air and earth*	*Kinds*	*Mean hardness*	*Least*	*Greatest*
0	0	20	3.18	2.92	3.41
0	1	16	4.59	4.47	4.71
1	0	16	5.00	4.86	5.20
0	2	8	5.99	5.84	6.10
1	1	8	6.38	6.26	6.47
2	0	8	6.81	6.58	6.93

The bias does not change the hardness, but it makes a crystal brittle. Crystals with a great number of bias are mostly broken in the vein, and those that come out whole are small. The greatest pieces I have seen are these: with a number of bias of nought, four inches and six tenths; of one, four inches and one tenth; of three, three inches; of four, two inches and four tenths; of five, one inch and nine tenths; and of the eight kinds whose number is eight, the greatest is no more than three and a half tenths of an inch, like a grain of rice, and cracks at the least knock. How many pieces of each kind I have seen is set down in the records left with the Society. When these six are drawn together, as in Fig. 3, the nearer the number comes to eight and a half, the smaller the pieces; a straight line drawn through the six comes to nothing at some eight and a half, between 8.3 and 8.7 according to how it is drawn. This shows only that it agrees with the bound of the trials of added bias, and does not fix a bound of its own: the kinds with much bias are rare, and fewer seen, so that the greatest seen are smaller. I suppose that the larger a crystal grows, the harder it is for it to escape some disturbance somewhere, and the nearer its bias lies to the bound, the less disturbance is needed to crack it; and so it cannot grow large.

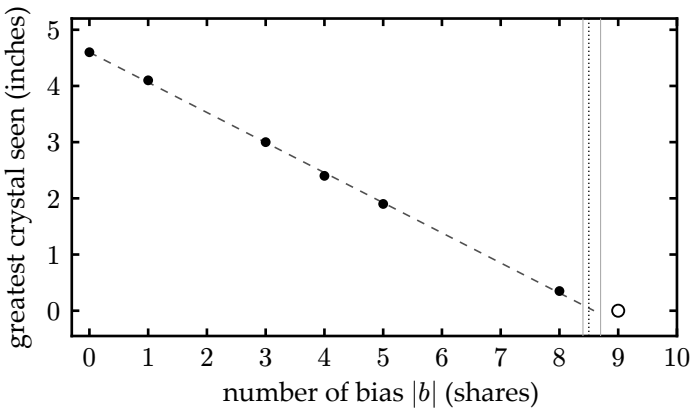


FIG. 3. The greatest crystal seen (inches) falls as the number of bias grows (filled points); the straight line drawn through the six comes to nothing at some eight and a half. The upright dotted line is the bound found by the trials of added bias in Table III, the thin lines beside it the least and greatest totals of the twenty-one pieces; the open point at the far right is the number of bias of the empty cells, nine

VII. SOME CORRECTIONS

When the table was laid out, several former opinions had to be changed.

First, what the craftsmen call the red-mist crystal and the flame-feather crystal both fall upon the point $(5, 0, 1, 0)$, the only one in its cell, and are alike in form, hardness and store: they are one crystal, from two places, the one from the eastern edge of the sea of clouds and the other from the veins of the sea of clouds worked from Kanaich.

Secondly, what is called the grey-sand crystal falls upon two points, $(1, 0, 1, 4)$ and $(0, 1, 1, 4)$, in two cells, with numbers of bias of contrary sign; their hardness and store are the same and both are slanting prisms, but slanting contrary ways, which no one had told apart before.

Thirdly, three crystals formerly measured by a workshop at Parin did not fall upon points; when I measured them again at Kanaich they all did. The former measurement had not taken off the background, and alchemy was worked in that workshop all day long.

Fourthly, nine pieces were sold as crystals of a single form, and their affinities did not fall upon points but upon the lines between two points; I cut them open, and all nine were twins.

Beyond these, I have set aside no crystal of a single form because its affinity did not agree. The table can therefore also be used to try a measurement: if the affinity of a crystal does not fall upon a point, it should first be asked whether the crystal is a twin, and then whether the instrument or the background is at fault.

VIII. THE FOUR EMPTY CELLS

The four cells at the middles of the sides of the table I leave empty. By the nomological algebra a crystal can form at each of their points, but it cannot stand

in ordinary places. If any place can bear more than nine shares of bias, they can grow there, and what they will be like can be written beforehand by the rules of the foregoing sections, as in Table VI.

Table VI. The crystals of the four empty cells, as the foregoing sections give them

Point	*Frame*	*Number of bias*	*Store (to the cubic inch)*	*Hardness*	*Form*
$(3, 0, 3, 0)$	$3F + 3G$	+9	3.6	about 3.2	Double pyramid
$(3, 0, 0, 3)$	$3F - 3G$	−9	3.6	about 3.2	Double pyramid
$(0, 3, 3, 0)$	$-3F + 3G$	−9	3.6	about 3.2	Double pyramid
$(0, 3, 0, 3)$	$-3F - 3G$	+9	3.6	about 3.2	Double pyramid

The affinity is the point itself. The store, by Table IV, with a frame of eighteen shares of strength, is some 3.6 units. The hardness, by Table V, since none of the four points has a cancelled pair, should be near 3.2. The form, by the fifth section, since all four points lie upon the two slanting lines through the empty cells, should be a double pyramid. Each of these points lies at the middle of the line between two common crystals, and an even twin falls there too, as a twin of pure fire crystal and pure air crystal in equal parts falls upon $(3, 0, 3, 0)$; whoever says he has found a crystal at these points must first cut it and see that it is of a single form, and then look to its form, hardness and store. None of these four crystals have I seen in any ordinary place of the world; the stations in the Belt of Rifted Law took more than twenty samples for me, and they were not there either. Where a place may be found that bears more than nine shares of bias, I do not know.

One thing more I can only suppose. In the trials of added bias, the bias was forced in step by step by an array, and when it passed the bound the whole piece cracked at once. If these four crystals grow elsewhere and are carried into an ordinary place, every lattice-cell stands only some half share above the bound, and that not forced in from without; I suppose that they will not break all at once, but will lose their form little by little, beginning at their faces. This has not been tried, and is only a supposition. To keep one, the vessel must bear for it at least the half share of bias that is over, and something to spare.

IX. CONCLUSION

From the foregoing sections I conclude:

1. The affinities of natural crystals of a single form all fall upon points a sixth apart; there are eighty-four such points within the tetrahedron.
2. By the nomological algebra of Elworthy and the trials of added bias, the lattice at a point has $xF + yG$ for its frame, and one lattice-cell brings forth xy shares of bias. The four points whose frame is the null can form no crystal; an ordinary place can bear only some eight and a half shares of bias, and the four points

whose number of bias is nine cannot stand in ordinary places; at the other seventy-six crystals can form in ordinary places, and all seventy-six I have found.
3. Crystals in one cell have the same frame, and the same store and form; the store stands in proportion to the strength of the frame, and the form follows only the way the frame points.
4. The hardness follows only which pairs cancel: every pair of fire and water adds some 1.8, every pair of air and earth some 1.4.
5. Crystals with a great number of bias are brittle and do not grow large; the greatest crystal seen comes to nothing at some eight and a half, which agrees with the bound found by the trials of added bias.
6. Where an affinity does not fall upon a point, the sample is twinned or mixed, or has been ill measured.
7. The affinity, store, hardness and form of the four empty cells can all be written beforehand by the foregoing sections; they can stand only where more than nine shares of bias can be borne. That if carried into an ordinary place they will slowly lose their form is my supposition.

There are several things of which I cannot give the cause: why the points lie a sixth apart; why, where the lines of the two axes cross, the axis of fire and water always comes first; why the cancelled pairs make a crystal harder, and why a pair of fire and water more than a pair of air and earth; and why an ordinary place bears no more than eight and a half shares of bias. All that is said here, moreover, holds only at the common warmth of a room, in ordinary places and in ordinary years: in another kind of place, if the bias it can bear is not the same, the points of the table that can form crystals will not be the same either. Of the seventy-six crystals I have measured I have left one piece of each, together with the records of the measurements, with the Silver Society.